Genius Invokation TCG Reinforcement-learning statistical report

279,573 completed games | 601 updates | 20,056,733 learner decisions

Assessment

The agent has learned substantial, measurable playing skill. It now reliably beats the scripted heuristic within this training distribution, and nearly always beats random play when its side has a listed 30-card deck. Improvement continues beyond the middle of training, but the final 101 updates show little further gain against the heuristic.

Latest training window: updates 551-601	Wins / games	Win rate
Heuristic	3,946 / 4,960	79.6%
Random	1,238 / 1,262	98.1%
Snapshot	2,686 / 4,919	54.6%

What matters most

Learning is real; recent gains are modest. Heuristic win rate rose from 23.3% at updates 1-50 to 73.5% at 251-300 and 79.6% at 551-601. The last two windows are effectively flat, with uncertainty still allowing a small gain or loss.

Unusual stage sides dominate residual random losses. At updates 501-601, 47 of 53 losses to random occurred on learner sides without listed action cards. Their win rate was 61.2%, versus 99.7% for sides with a listed deck. These are different task conditions.

The opening value is not a calibrated opponent-specific win probability. In the last 101 updates it predicts about 57% across opponent types, despite actual rates ranging from about 50% in self-play to 98% against random.

No evidence here establishes human-level or deployment-level strength. This is one evolving training run, plus a repeatedly reused 100-matchup benchmark. Human opponents, coherent held-out decks and real-game execution are not evaluated.

Prepared from all eight CSV files, the supplied README, and the owner's benchmark/training clarifications. No external game balance assumptions are used. Names are taken from definitions.csv.

Study design and interpretation

What was trained

The supplied protocol describes a 1.2-million-weight PPO network with terminal rewards of +1 for a win and -1 for a loss. A round-15 limit is a loss for both players. Decks come from the same 512 simulated collections and a stable mixture of collection decks, NPC decks and PvE stages. The deck builder itself is not trained.

Component	Protocol and consequence
Opponents	55% self-play, 20% rolling snapshots, 20% heuristic, 5% random. Before snapshots existed, their allocation went to self-play.
Snapshot pool	Saved every 10 updates; at most 20 snapshots. Each of four workers receives one uniformly selected snapshot per update. Results share worker/opponent context.
Benchmark	100 fixed matchups, each played with the learner in both seats. Same decks, flags, equipment and game seeds on every evaluation. 200 nominal games per opponent.
Action selection	Learner and snapshots sample actions. Benchmark outcomes are therefore stochastic, even with the same fixed game seeds.
Timeline	An update label records when a game finished. A game can contain decisions from more than one policy version. These are training-window outcomes, not pure checkpoint tests.

Populations are kept separate

Non-self win rates use one learner seat per game, identified from the controller columns. All completed games remain in the denominator, including round-limit losses. **Self-play** supplies useful behavior and value data but cannot show absolute strength: one learner usually wins and one loses.

Primary phases: early = updates 1-50; middle = 251-300; final = 551-601. Larger late-period slices use 501-601 for subgroup precision and 401-601 for exploratory character/card associations. Every table identifies its window.

Confidence intervals quantify variation within this run, not variation across independent training seeds. Training-window intervals use resampled five-update blocks where indicated; benchmark aggregates lack the match-level data required for a valid paired analysis.

Data coverage and integrity

Source file	Rows	Role
games.csv	279,573	Game outcomes and configuration
seat_summary.csv	559,146	Two seat summaries per game
game_characters.csv	1,666,947	Starting character rosters
game_equipment.csv	12,836	Pre-equipped definitions
moves_sample.csv	461,008	Complete traces for 5,000 games
training_metrics.csv	601	One row per update
definitions.csv	3,588	Names and object types
game_cards.csv	12,145,732	Starting action-card entries

Reconciliation checks passed

Game keys are unique. Every game has exactly two unique seat summaries. Their move totals equal game move_count for every game. The 5,000 sampled traces have contiguous move numbers and exactly the recorded number of moves. Training win rates reconstructed from completed games match every reported opponent-specific rate to floating-point precision.

The seat summaries contain **25,746,860 total decisions**, of which **20,056,733 are learner decisions**. That learner total exactly matches the final cumulative learner_steps metric. All referenced character and card IDs resolve in definitions.csv; no duplicate card keys or character slots were found.

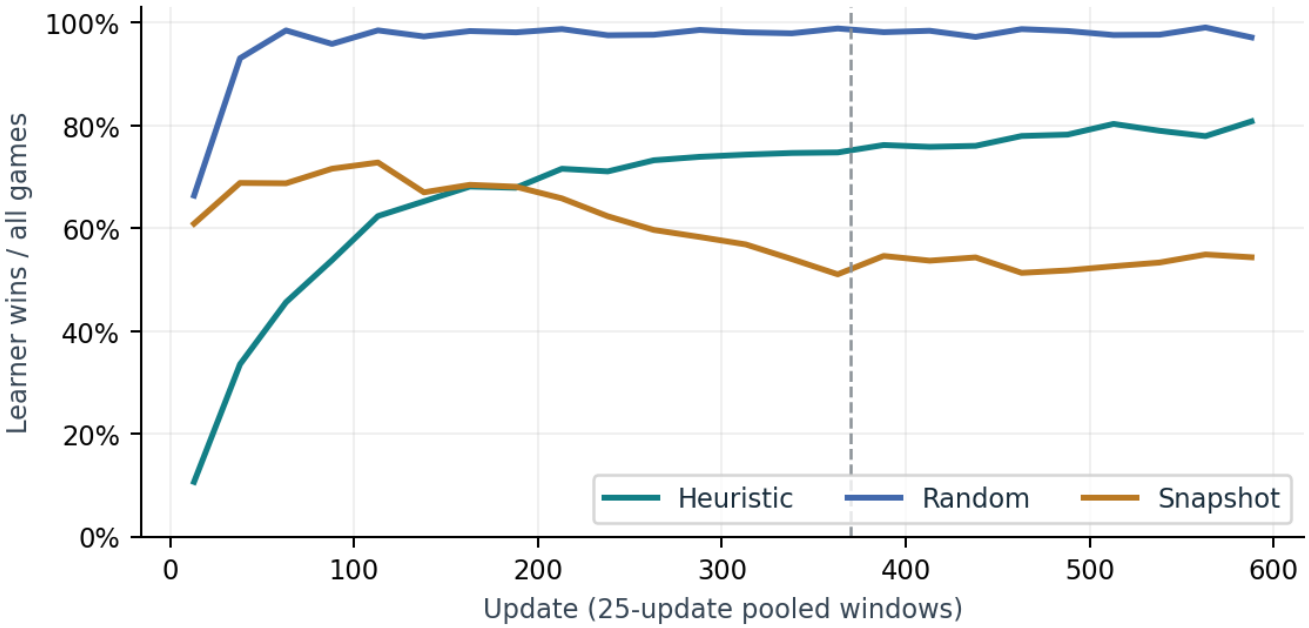
Conventions that matter

All 1,512 missing winners occur at round 15 and are counted as losses. Starting rosters are not universally three characters: some stage sides have one, two or four. Checking defeated counts against each actual roster resolves all apparent outcome inconsistencies.

There are 21,747 seat-games without action-card rows; all are in stage games. Every seat with a listed action deck has exactly 30 cards. Absence of rows is treated as “no listed cards,” not automatically as proof that the exporter captured every deck.

The archive contains 601 updates rather than the README's rounded description of 600. The first 370 learning-rate entries are missing; their constant value is supplied by the owner, not imputed from the CSV.

Learning progress across the run



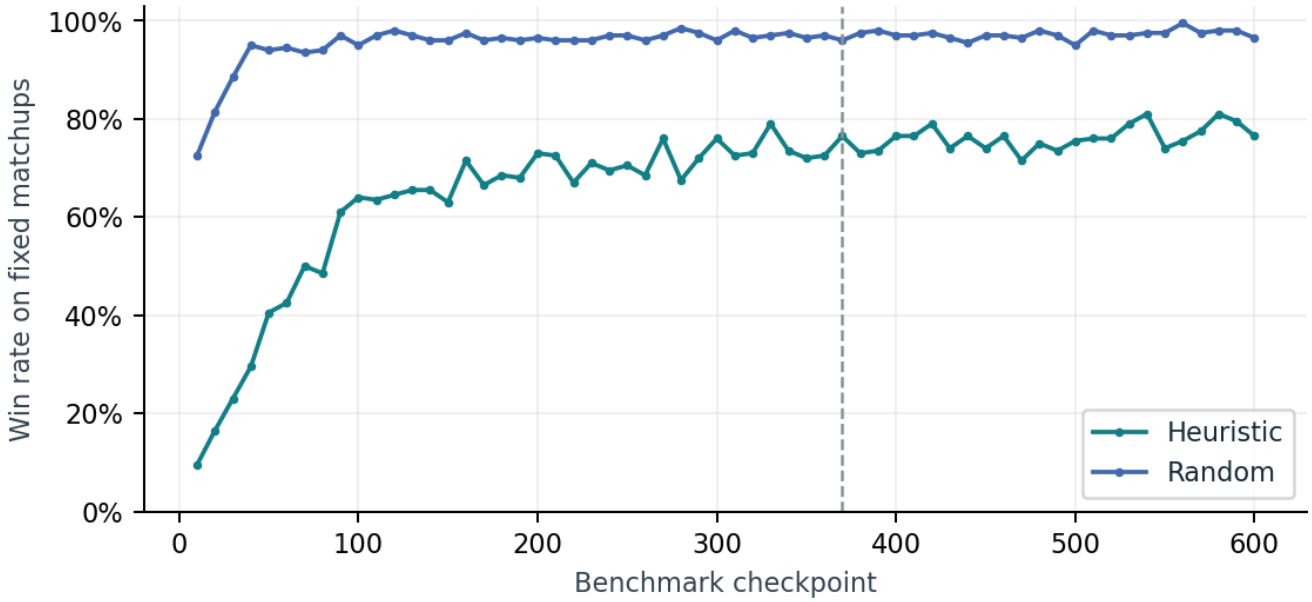
Window	Heuristic: wins / games	Win rate	95% block interval
1-50	732 / 3,137	23.3%	15.6% to 30.1%
51-100	2,113 / 4,249	49.7%	47.0% to 52.5%
251-300	3,657 / 4,974	73.5%	72.4% to 74.5%
321-370	3,602 / 4,855	74.2%	72.8% to 75.6%
371-420	3,813 / 4,998	76.3%	74.7% to 77.8%
501-550	3,889 / 4,887	79.6%	78.4% to 80.9%
551-601	3,946 / 4,960	79.6%	78.2% to 80.9%

The largest gains occur early. Random-opponent wins are already 97.1% at updates 51-100, while heuristic wins are still 49.7%. The heuristic therefore remains a more informative difficulty test later in training.

From the middle to the final window, heuristic performance improves by **+6.0 pp** (95% block interval +4.3 pp to +7.8 pp). From updates 501-550 to 551-601 the change is **-0.02 pp** (-1.9 pp to +1.8 pp). This supports a recent slowdown, not proof that further learning is impossible.

Source: games.csv and seat_summary.csv. Curves show complete 25-update bins through update 600; tables include update 601. Intervals in the table use circular five-update blocks and 5,000 bootstrap resamples. Early-window nonstationarity makes its interval especially broad.

The fixed benchmark



Checkpoints	Mean heuristic rate	Mean random rate
10-50	23.8%	86.3%
260-300	72.0%	97.0%
320-370	74.4%	96.7%
380-420	75.7%	97.4%
510-550	77.2%	97.4%
560-600	78.0%	97.9%

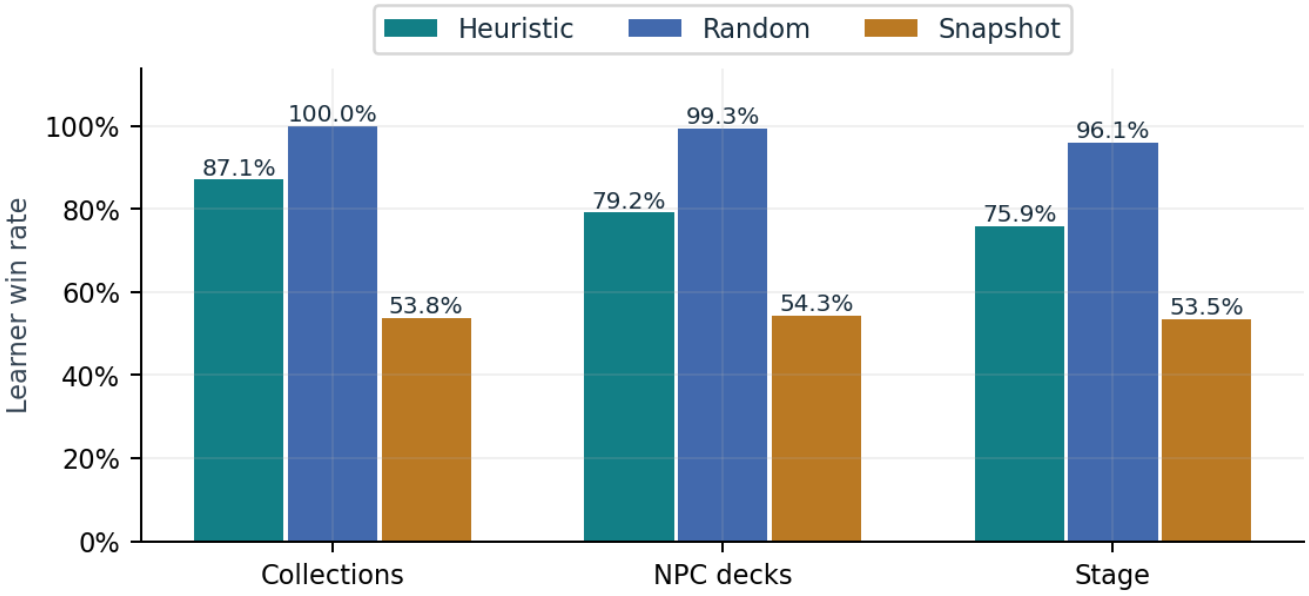
The final checkpoint (600) scores **153/200 against the heuristic (76.5%)** and **193/200 against random (96.5%)**. Heuristic scores reach 81.0% at both updates 540 and 580. Choosing the largest observed score would favor favorable sampling noise as well as skill.

A hypothetical independent 200-game binomial interval for 153 wins is about 70.1%-81.8%. It is only a scale reference: the actual design has 100 paired matchups, recurring across checkpoints. Without per-matchup outcomes, pair dependence and cross-checkpoint covariance cannot be estimated, so no formal claim of a 540-to-600 regression is warranted.

The benchmark stays comparable across the restart because the fixed matchup list was verified byte-identical. Its sampled policy still creates rerun noise. Checkpoint averages above are descriptive summaries of the same recurring tasks, not larger independent test sets.

One logging exception: random score at update 510 is 0.9798994975, exactly 195/199, rather than a multiple of 1/200. This is consistent with an omitted or failed game but does not establish the cause. Retain the reported rate; log attempted, completed and failed counts in future evaluations.

Deck source and stage-side effects



At updates 501-601, heuristic win rate is 87.1% for collections, 79.2% for NPC decks and 75.9% for stages. These source groups mix different rosters and deck conditions, so the gap is not a clean estimate of tactical difficulty alone.

Opponent	Learner has listed cards	Wins / games	Win rate
Heuristic	No card rows	52 / 399	13.0%
Heuristic	Yes (30 cards)	7,783 / 9,448	82.4%
Random	No card rows	74 / 121	61.2%
Random	Yes (30 cards)	2,335 / 2,341	99.7%
Snapshot	No card rows	24 / 402	6.0%
Snapshot	Yes (30 cards)	5,283 / 9,469	55.8%

The no-listed-card group is small but disproportionately hard. It contributes 347 of 2,012 heuristic losses (17.2%) and 47 of 53 random losses (88.7%). For random opponents, the remaining listed-deck group wins 2,335/2,341 games, or 99.74%.

This finding is consistent with special NPC sides having very different capabilities; it does not identify the missing-card group as an engine defect or justify excluding it from the headline score. Explicitly recording the stage role and whether the side is playable in the intended deployment would settle which metric matters most.

Source: updates 501-601; game_cards.csv joined by game_key and seat. “No card rows” is an export observation. Source labels are game-level and do not identify the NPC side.

Seat position and deck knowledge

Opponent, updates 501-601	Seat 0	Seat 1	Knows deck	Blind
Heuristic	79.6%	79.5%	79.0%	80.3%
Random	98.1%	97.6%	98.7%	96.7%
Snapshot	54.8%	52.7%	54.4%	52.8%

Raw differences are small or inconsistent across opponent types. Against the heuristic, known-deck seats actually have a slightly lower observed rate, but their task mix differs. That does not show that information harms the policy.

Adjusted descriptive comparison

An additive linear probability model uses 44,057 non-self games from updates 401-601. It controls for opponent, source, both knowledge flags, learner seat and linear update trend. Intervals cluster residual variation by finishing update.

Term	Adjusted difference	95% interval
NPC decks vs collections	-4.03 pp	-5.20 pp to -2.87 pp
Stage vs collections	-5.45 pp	-6.48 pp to -4.43 pp
Seat 1 vs seat 0	-0.34 pp	-1.12 pp to +0.45 pp
Learner knows opponent deck	+0.28 pp	-0.48 pp to +1.05 pp
Opponent knows learner deck	+0.02 pp	-0.83 pp to +0.87 pp
Per 100 additional updates	+1.32 pp	+0.61 pp to +2.02 pp

There is no clear pooled seat advantage or deck-knowledge benefit in this specification. The estimated learner knowledge difference is +0.28 percentage points, with a range from -0.48 to +1.05. This cannot establish that the network ignores the input: benefits may depend on particular matchups or game states.

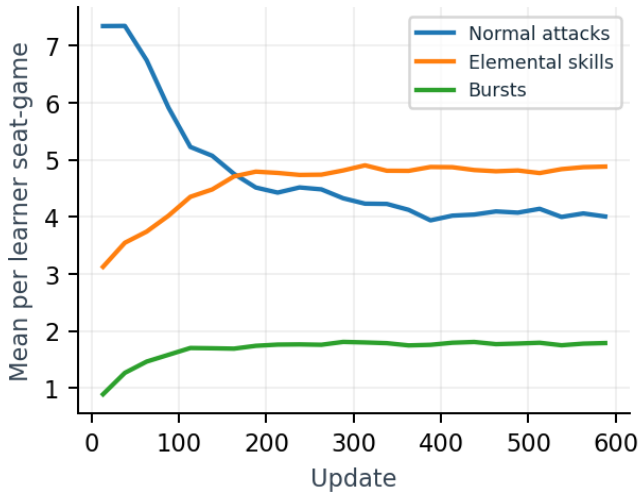
The source penalties remain after these basic controls. Character composition, actual stage role, no-listed-card status and opponent snapshot identity are not controlled, so this is not a causal model. Common additive effects can also hide opponent-specific differences.

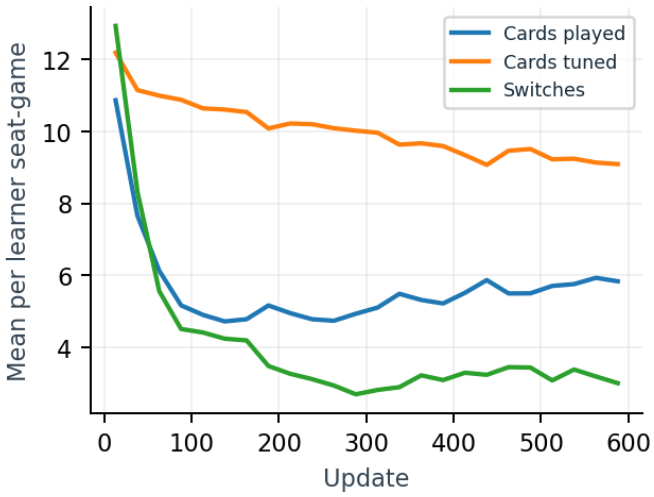
A more decisive information test

For a fixed checkpoint, evaluate the same matchups and seats with only the knowledge flag toggled, using repeated policy samples. Pair the outcomes by matchup. This directly tests whether the information helps under otherwise comparable conditions.

Source: games.csv and seat_summary.csv. This model is used to summarize associations, not to generate bounded per-game probability forecasts.

How the playing style changed





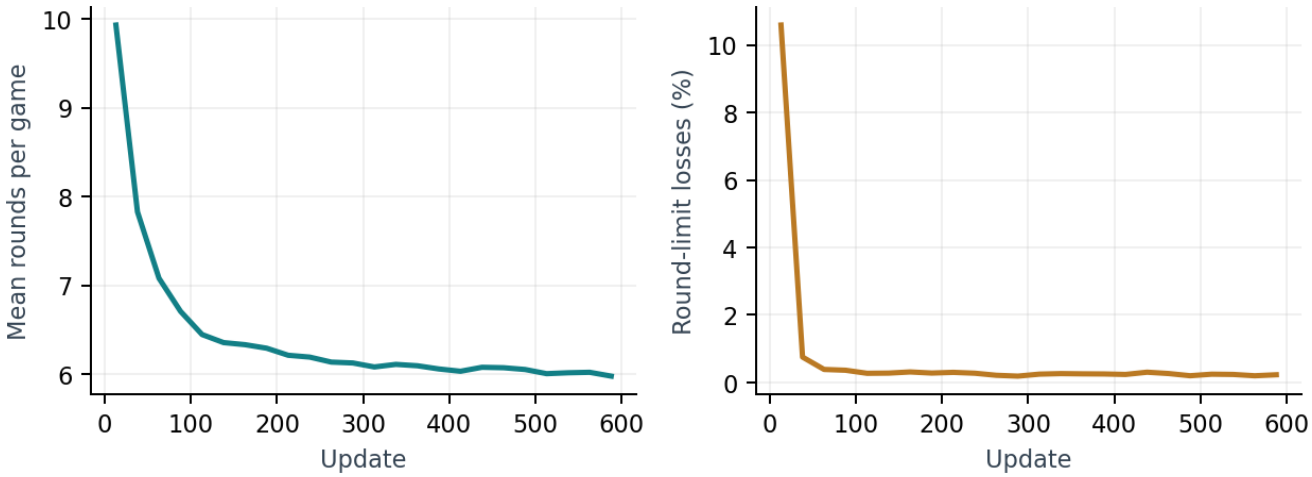
Learner metric	Early	Middle	Final
Normal attacks / game	7.35	4.41	4.04
Elemental skills / game	3.36	4.78	4.88
Bursts / game	1.10	1.79	1.79
Cards played / game	9.10	4.84	5.89
Cards tuned / game	11.62	10.05	9.11
Voluntary switches / game	10.40	2.82	3.10
Skills / round	1.35	1.82	1.82
Tuned share of played + tuned cards	56.1%	67.5%	60.7%

The agent moves from frequent normal attacks and switching toward elemental skills and bursts. Voluntary switches fall from 10.40 to 3.10 per seat-game, while skills per round rise from 1.35 to 1.82. Those shifts are consistent with more directed play, although damage and dice efficiency were not recorded.

Card use does not simply decline throughout training. Played cards fall in the middle, then recover to 5.89 per game. Tuning absorbs 60.7% of recorded played-or-tuned card consumption in the final window, down from 67.5% in the middle. Random, incoherent decks make this an ambiguous efficiency signal.

Source: all learner seats, including both self-play seats. Curves show 25-update bins through update 600. Per-game metrics average seat-games. Per-round rates divide pooled action counts by pooled game rounds, not by the number of turns the seat was active.

Game length and failure patterns



Mean game length falls from 8.75 rounds early to 6.00 in the final window, and total decisions fall from 125.38 to 87.76 per game. Round-limit losses fall from 5.06% to 0.209%, a roughly 96% relative reduction. Across the full run, median length is six rounds; 95% finish by round 10.

Late heuristic games (501-601)	Wins: n = 7,835	Losses: n = 2,012
Mean rounds	6.03	7.18
Mean voluntary switches	2.85	4.53
Mean bursts	2.22	1.77
Mean cards played	5.71	5.91
Mean skills used	11.83	11.67

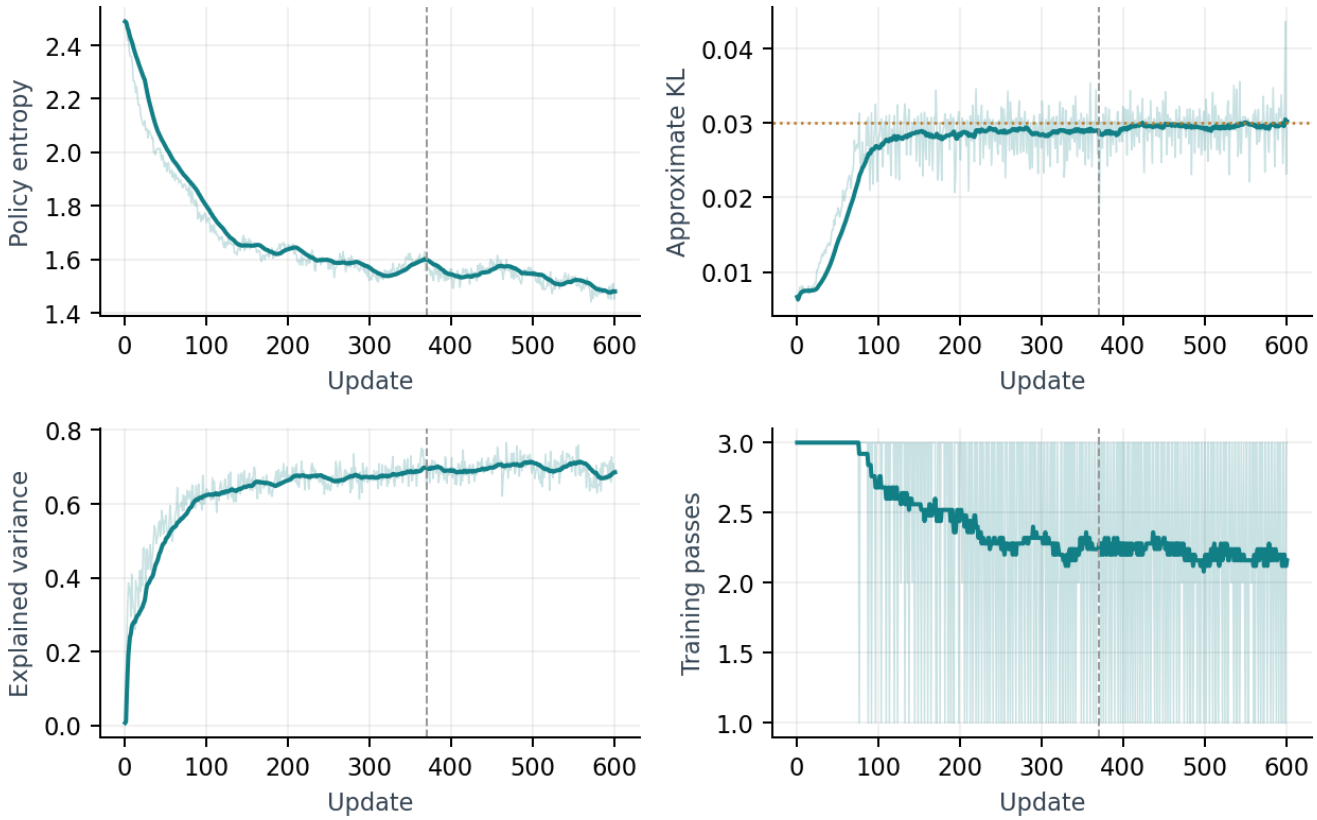
Heuristic losses take longer and contain more switching but fewer bursts. Similar total skill counts suggest that simply counting attacks will not explain the remaining losses. This is outcome-associated behavior: losing positions can force switches, so reducing switching mechanically could worsen play.

Concrete cases for replay inspection

Game key	Learner side	Observed result
503:104	Stage; 4 characters; no listed cards	Round-15 loss; 27 voluntary switches
501:158	Stage; 2 characters; no listed cards	Round-11 loss; 10 voluntary switches
502:374	Stage; 4 characters; no listed cards	Round-6 loss; 6 voluntary switches

Curves show 25-update bins through update 600. These are late losses against random, selected as examples rather than a representative sample. Full board states, damage, dice costs and counterfactual legal actions are absent; the records locate cases for debugging but cannot establish which move caused each loss.

PPO optimization and stability



Mean metric	Updates 1-50	251-300	551-601
Entropy	2.147	1.578	1.489
Approximate KL	0.011	0.029	0.030
Clip fraction	0.143	0.264	0.250
Explained variance	0.418	0.672	0.686
Training passes	3.000	2.300	2.196

Entropy declines, explained variance improves and then levels near 0.69, and approximate KL remains close to the 0.03 early-stop threshold. This shows active learning with a frequently binding update-size safeguard; it does not alone establish instability or a defective learning rate.

In the final 51 updates, 17 run one pass, seven run two and 27 run all three. Thus **33.3% use one pass**, while **47.1% stop below three**. Reported KL exceeds 0.03 in 27 updates; an early-stop threshold is not necessarily a hard upper bound on the final logged statistic.

Thin lines are update-level values; thick lines are trailing 25-update means. The dashed vertical line marks the restart. Value loss is about 0.052 on average in both the middle and final windows; its training-target scale should not be confused with probability calibration.

Snapshots and the update-370 restart

A moving opponent changes the meaning of the score

Snapshot win rate is 70.1% at updates 51-100, 58.9% at 251-300 and 54.6% at 551-601. The oldest easy opponents continually leave the pool. A lower rate against this improving pool does not imply the agent has become weaker.

The final-window snapshot advantage is 4.6 percentage points above 50%; its five-update block interval is 53.5% to 55.7%. This is evidence of an advantage over the sampled recent pool within this run, not an Elo estimate or proof of improvement against every historical checkpoint.

Before / after restart	Updates 321-370	371-420
Heuristic training win rate	74.2%	76.3%
Snapshot training win rate	52.6%	54.3%
Random training win rate	98.2%	98.3%
Mean approximate KL	0.0291	0.0291
Mean training passes	2.22	2.28
Mean explained variance	0.690	0.689

What can be concluded about the restart

There is no large immediate discontinuity in these summary windows. Learning rate fell from the prior 0.000200 setting to the resumed decay schedule near 0.000172, reaching about 0.000155 by the end. The missing early CSV values are covered by the owner's protocol.

The restart also fixed the rare Lisa pre-equipped-talent crash, reset random streams, discarded unfinished games and replayed their updates. Without a control run, the post-restart improvement cannot be attributed specifically to learning-rate decay. Excluded crashes also make this a completed-game analysis rather than an attempted-game analysis.

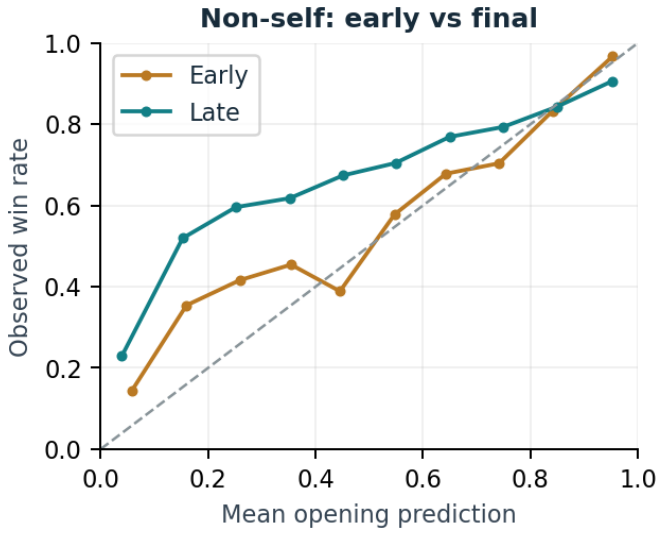
There are 8,164 repeated seed entries beyond their first occurrence. Repeated seeds are compatible with the restarted streams and need not produce repeated games. Only 15 records have identical logged configurations and outcomes after ignoring game key, update and policy version; that still does not prove duplicate trajectories.

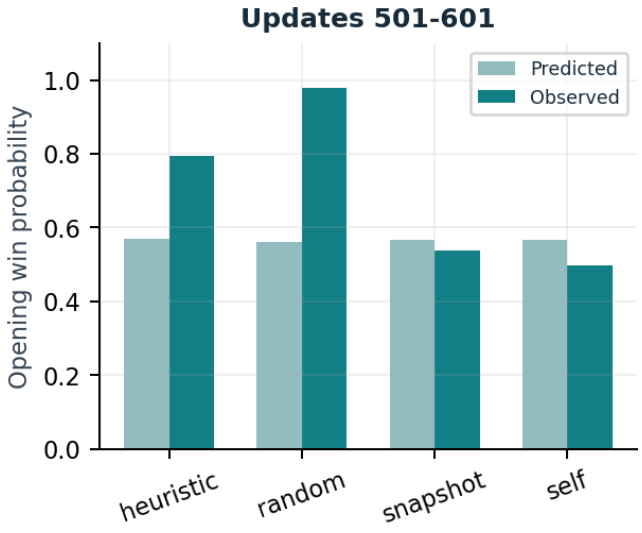
What would resolve snapshot progress

Save the selected snapshot version and worker ID on each game, then evaluate chosen checkpoints against a fixed historical panel. That would separate opponent age from learning progress and expose possible forgetting or cyclic matchups.

Source: user protocol, games.csv and training_metrics.csv. Policy-version gaps are not interpreted as exact within-game staleness: games can span multiple versions.

Value estimates: useful ranking, weak calibration





Opponent, updates 501-601	Opening prediction	Actual wins	Opening Brier	Final Brier
Heuristic	56.9%	79.6%	0.213	0.065
Random	56.1%	97.8%	0.252	0.010
Snapshot	56.8%	53.8%	0.207	0.075
Self	56.7%	49.9%	0.214	0.077

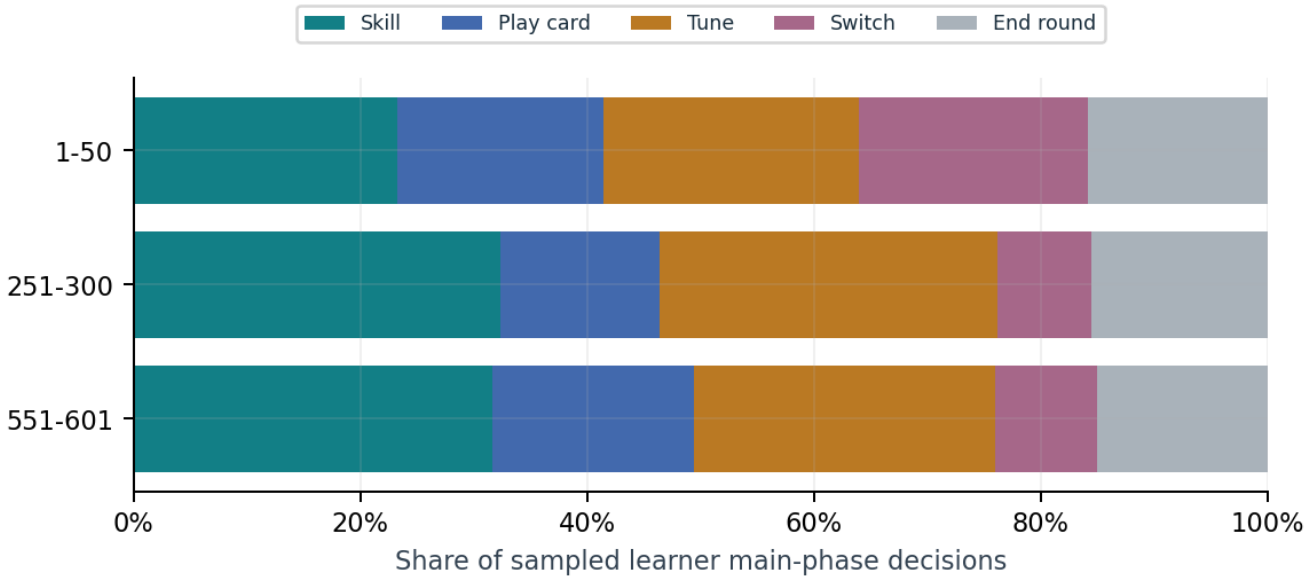
The opening estimate clusters around 56%-57% on average across all opponent types. Against the heuristic it underestimates wins by 22.7 percentage points; against random by 41.8 points. In self-play it overestimates by 6.8 points. A shared value function may reflect the training mixture or omit opponent identity; the export does not specify its inputs.

It still ranks positions meaningfully: opening AUC is 0.731 against the heuristic and 0.746 against snapshots (0.5 means chance ranking). Ranking and calibration are different. The last-decision heuristic AUC improves to 0.955, and its Brier score falls to 0.065.

For final-window non-self games, opening Brier is 0.216, worse than the same-sample constant-rate baseline of 0.207; last-decision Brier is 0.064. A later prediction sees much more of the game, so this comparison is not evidence of opening forecasting quality.

The transformation p = (value + 1) / 2 follows the supplied README. It is only literally a win probability if value estimates the undiscounted expected terminal reward with no incompatible target transformation. Discount/target details were not supplied. Reliability curve: non-self early vs final windows, ten equal-width bins.

Action selection and card usage



The 5,000-game sample contains 461,008 decisions, including 361,139 learner decisions. In the final window, skills take 31.6% of sampled main-phase choices, tuning 26.6%, card play 17.8%, ending the round 15.0% and switching 9.0%. These are decision shares, not dice-spend shares.

Most played cards in sampled learner moves	Plays	Distinct games
The Bestest Travel Companion!	2,777	2,061
Favonius Cathedral	1,475	1,275
Strategize	1,434	1,170
Sweet Madame	1,179	1,022
Mondstadt Hash Brown	1,158	929
Paimon	1,059	951
Changing Shifts	817	731
Timmie	770	631

Strategize is also the most frequently tuned card in the sample (2,003 times), followed by Dawn Winery (1,822) and Changing Shifts (1,609). A card may be useful in one hand and appropriate to tune in another. Counts alone cannot determine whether tuning decisions are good; hand availability and costs are missing.

Entropy needs a decision-specific denominator

For main-phase decisions with more than one option, mean entropy divided by log(option_count) falls from 0.922 early to 0.591 in the final window. The policy becomes more selective but remains stochastic. These are move-weighted sample diagnostics.

Reroll and mulligan entropy often exceeds log(option_count), while staying within option_count × log(2). That is consistent with multiple binary choices rather than a single categorical action; it is not evidence of an invalid probability distribution. Use that interpretation provisionally until the logging implementation is confirmed.

Character associations worth investigating

These rankings use 19,513 learner games against the heuristic at updates 401-601. A character needs at least 200 appearances; 79 qualify. “Adjusted residual” is the mean outcome minus the win rate for that source, learner seat and 25-update bin. It partially removes task mix and time, but not teammates, cards or opposing roster.

Higher residual associations	Games	Raw wins	Residual
Ineffa	416	88.9%	+12.1 pp
Thoma	394	90.4%	+12.1 pp
Ganyu	623	88.6%	+11.6 pp
Qiqi	355	86.8%	+10.0 pp
Hu Tao	319	86.5%	+9.8 pp
Diona	511	86.1%	+9.6 pp
Cyno	582	86.3%	+9.5 pp
Kamisato Ayato	359	87.2%	+9.5 pp

Lower residual associations	Games	Raw wins	Residual
Yaoyao	216	68.5%	-9.0 pp
Barbara	457	70.2%	-7.0 pp
Unresolved name (ID 1506)	201	72.6%	-6.7 pp
Sethos	252	73.0%	-6.1 pp
Sigewinne	257	73.2%	-5.1 pp
Xilonen	321	73.2%	-4.9 pp
Citlali	383	73.9%	-4.4 pp
Escoffier	261	74.3%	-4.0 pp

These are associations, not a character tier list. Ineffa, Thoma and Ganyu occur in successful configurations, while Yaoyao and Barbara appear in weaker configurations under this policy and generator. Different cards, teammate combinations, NPC templates and opposing decks can explain part or all of a gap.

Raw rates with a few hundred games are noisy. For example, Thoma's 356/394 rate is 90.4%, with an ordinary descriptive Wilson interval of approximately 87.0%-92.9%; this ignores shared update and deck dependence. Selecting extremes among 79 characters further exaggerates apparent gaps.

The appropriate follow-up is controlled deck-swap evaluation with a fixed checkpoint and matched opponents. Do not directly train the deck builder on these ranks as if they represented isolated character value. The unresolved template name in definitions.csv is preserved as ID 1506 rather than guessed.

Action-card inclusion associations

The same late heuristic population is used here. Each card counts once per seat-game where it appears in the starting list, regardless of copies; at least 500 appearances are required. There are 172 eligible cards out of 481 observed IDs. Residual adjustment again uses source, seat and 25-update bin.

Higher inclusion associations	Games	Raw wins	Residual
Grand Narukami Shrine	825	85.2%	+7.9 pp
Elemental Resonance: Woven Ice	826	83.8%	+7.7 pp
Elemental Resonance: Woven Stone	606	83.7%	+7.6 pp
Elemental Resonance: Shattering Ice	869	83.5%	+7.4 pp
Elemental Resonance: Enduring Rock	591	83.2%	+7.2 pp
Sangonomiya Shrine	883	84.7%	+7.1 pp
Sacrificial Bow	856	84.1%	+6.8 pp
Jade Chamber	721	83.9%	+6.3 pp

Lower inclusion associations	Games	Raw wins	Residual
Atea	521	69.1%	-9.7 pp
Heart of Depth	612	72.9%	-6.0 pp
"An Adventure Through the Morning Mist"	502	75.1%	-5.6 pp
Sapwood Blade	536	72.9%	-5.5 pp
Maiden's Fading Beauty	602	73.6%	-5.3 pp
Cleaning Time	562	73.8%	-5.1 pp
The Boar Princess	559	74.2%	-5.1 pp
Saurian Dining Buddies	526	75.3%	-4.7 pp

Grand Narukami Shrine has a +7.9-point average residual, and Atea a -9.7-point residual. These are conditional descriptions of generated decks, not the effect of adding or removing those cards.

Card inclusion can identify roster families rather than isolated card quality. The resonance entries above are candidates for this kind of confounding. Equipment availability, collection ownership and NPC templates likewise influence card inclusion. A deck may win without drawing or playing the listed card at all.

Ranks were selected from many candidates and have no multiplicity-adjusted significance claim. Treat them as priorities for matched card-replacement experiments. Full hand, draw and legal-play logging would distinguish useful cards from cards merely attached to successful decks.

Roster coverage and repeated configurations

Starting characters per seat	Seat-games	Share
1	5,485	0.98%
2	4,645	0.83%
3	543,892	97.27%
4	5,124	0.92%

All non-three-character rosters occur in stage games. There are 260 distinct character IDs across playable and NPC definitions, 481 action-card IDs, and 12,836 games with pre-equipped equipment. All listed action decks total 30 cards, but 3.89% of seat-games have no card entries.

Frequent late heuristic character lineups

Sorted lineup (order ignored)	Games	Win rate
Kaeya / Diluc / Fischl	741	86.6%
Kaeya / Diluc / Sucrose	675	79.6%
Kaeya / Fischl / Sucrose	668	88.5%
Diluc / Fischl / Sucrose	668	88.0%
Mona / Cyno / Ineffa	100	91.0%
Sangonomiya Kokomi / Dehya / Ningguang	93	84.9%

There are 4,998 distinct sorted lineups in the late heuristic slice, but the four most frequent account for 2,752 games (14.1%). Many lineups therefore have little evidence, while a few receive repeated exposure. These are character sets, not complete 30-card deck identities.

The large number of games should not be read as an equally large number of independent deck environments. The same 512 collections, NPC templates and benchmark list recur. Distinct random seeds diversify trajectories but do not by themselves establish generalization to new collections or coherent player-built decks.

What to log for deck-builder research

Record collection ID, exact deck hash, NPC/stage identity, stage-side role, roster type and pre-equipped state. Reserve unseen collection IDs and deck families before tuning a deck builder. Otherwise the play policy and builder can appear strong by specializing in the same recurring combinations.

Source: game_characters.csv, game_cards.csv, game_equipment.csv and definitions.csv. Counts refer to the full run unless a late heuristic slice is explicitly named.

Priorities for the next training cycle

Priority	Concrete action	Why this run supports it
1. Separate deployment tasks	Log stage identity and learner role; distinguish intended player decks from special NPC sides. Keep the current mixed score too.	No-listed-card sides account for 88.7% of late random losses and 17.2% of heuristic losses.
2. Make evaluation auditable	Save each benchmark matchup, seat, seed, outcome, checkpoint and failure. Repeat shortlisted checkpoints with independent policy samples.	Only aggregate rates survive; paired uncertainty and the 199-game exception cannot be resolved.
3. Add fresh held-out decks	Retain the fixed benchmark for continuity, plus a locked fresh test set from unseen deck/collection families.	The same 100 matchups recur and the generator reuses 512 collections.
4. Inspect concentrated failures	Replay the no-listed-card cases and later heuristic losses with high switching or limited bursts; log dice paid and damage.	Losses show identifiable patterns, but the export cannot distinguish forced responses from poor choices.
5. Validate the value head	Confirm discount/target semantics and opponent inputs; measure calibration by opponent and game stage on held-out data.	Opening values barely reflect large differences in opponent strength.
6. Test changes one at a time	Compare learning-rate or update-pass settings from the same checkpoint with matched evaluation and multiple training seeds.	The KL safeguard frequently limits passes, while late gains slow; the restart confounds causal attribution.

How to choose the next checkpoint

Do not choose solely on the maximum observed benchmark score. Shortlist late checkpoints using the trend, then rerun them on matched conditions and compare paired results. For player-deck deployment, report both the relevant player-side subset and the full existing mixed test.

A 200-game benchmark is useful for large changes but too noisy for confident 1-2-point decisions without repeated or broader evaluation. More repetitions on fixed tasks reduce action-sampling uncertainty; more distinct matchups improve coverage. Those solve different problems.

These are proposed experiments, not claims that a particular optimizer change, reward redesign or card choice will improve the bot. No such intervention was evaluated in this archive.

Methods, limitations and provenance

Definitions and computation

Win: winner equals the learner seat. A missing winner contributes zero wins and one game. **Rate:** total wins / total games, not an average of per-update rates. **Self-play:** both learner seats are retained only for explicitly labeled behavior/value analyses. No absolute-strength claim uses its near-50% seat win rate.

Block intervals: aggregate wins and games by finishing update, resample circular blocks of five consecutive updates within a window, concatenate to the original number of updates, then recompute the ratio. The 2.5th and 97.5th percentiles of 5,000 resamples form the interval. Window differences resample each window separately. This allows short local dependence, not arbitrary long-range dependence or training-run variability.

Adjusted model: ordinary least squares on binary wins with an intercept, opponent/source indicators, both knowledge flags, seat and centered linear update. Covariance uses update-cluster score sums with finite-cluster correction; intervals are estimate ±1.96 standard errors. No character/card effect is inferred causally from it.

Value diagnostics: p = (v + 1)/2, Brier = mean((p - win)^2), lower is better. Calibration compares observed rates to predicted probabilities; ECE uses ten equal-width bins. AUC uses ranked probabilities. Last_value is the last recorded decision estimate, not necessarily the terminal state. Baselines use the same sample's observed win prevalence and are descriptive, not out-of-sample fitted models.

Character/card residual: for each game subtract the mean win rate of its source × learner seat × 25-update stratum, then average among games containing the object. These are inclusive-stratum residuals, not present-versus-absent treatment effects. Candidate filtering and ranking are exploratory; no corrected significance tests are claimed.

Limits that remain

One training seed; adaptive and partially shared policies; missing attempted/crashed games; no snapshot IDs or worker IDs; no complete benchmark outcomes; no full board states, damage or dice expenditure; no draw/hand exposure data; no explicit stage-side label. Printed values are rounded, while calculations use the CSV precision. Tiny chosen probabilities can round to zero, so log-loss on recorded action probabilities was not used.

Sources and reproducibility

Primary source: external_analysis(1).zip, containing README.md and all eight named CSV files. Protocol source: owner clarifications supplied in this conversation on 13 October 2026, including fixed paired benchmarks, sampled actions, rolling snapshots, the update-370 restart and KL early stopping. No external balance rankings or unverified game rules were added.

Processing: Python/pandas/NumPy for joins and summaries; deterministic bootstrap seed 11013; SciPy ranked statistics; Matplotlib figures; ReportLab PDF. All full-population analyses use the complete relevant CSV files. Only action-level analyses use the provided 5,000-game sample. Original files were not modified.